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Task Description:

Space flight causes lymphopenia, abnormally low mitogen responses, and reduced production of antibodies and cytokines. Space flight-associated factors such as microgravity, lack of load-bearing, stress, and radiation are believed to contribute to these effects, but the exact mechanisms are unknown. Ground-based mouse models such as hindlimb unloading (HU) and irradiation cause similar effects. We hypothesize that these changes result from lymphocyte apoptosis induced by the cell surface receptor, Fas, in a manner dependent on endogenous opioids. Furthermore, we predict that radiation exposure and HU act synergistically. We have proposed to: 1) Elucidate the role of Fas and endogenous opioids in the modulation of the immune system by HU and radiation; 2) Determine the effect of HU and radiation on lymphocyte dynamics and immune responses, and 3) because autoimmune disease, paradoxically, is often associated with stress and radiation exposure, we will determine whether HU and radiation promote autoimmunity. The overall goal is to determine the mechanisms underlying apoptosis-related immune modulation using ground-based models.

PROGRESS

1. In a study at Brookhaven National Laboratory (BNL), we examined the acute effects of whole body HZE-particle irradiation. Mice were subjected to high-energy 56Fe ions and the effects on various lymphocyte populations were determined. We found lymphocyte in the spleen and thymus to be reduced in a dose-dependent manner (200, 100 and 50 cGy), with thymocytes more sensitive than splenocytes. Subpopulation analysis revealed that all subsets declined in numbers, except for CD4+CD25+ regulatory T cells. To examine the mechanism, mice deficient in apoptosis-inducing genes were tested: deficiency in Fas ligand (gld mutation) or TRAIL (TNF-related apoptosis-inducing ligand) did not prevent lymphocyte losses, indicating that these apoptosis pathways are not involved in particle radiation-induced lymphocyte losses. In contrast, gamma-radiation-induced cell depletion requires FasL and TRAIL. Thus, there are divergent pathways of apoptosis induction by particle radiation vs. gamma-radiation. Using specific inhibitors in vitro, we also found that particle radiation-induced lymphocyte apoptosis is caspase-dependent, but does not utilize nitric oxide or oxygen free radicals.

2. We have reported that endogenous granzyme B (GzmB) is a critical regulator of the balance between Th1/Th2 helper T cells. Highly expressed in Th2 cells, GzmB is required to induce apoptosis in these cells. We have found that GzmB-deficient mice develop stronger Th2-type immune responses. Since GzmB expression can be inhibited by steroid hormones, and corticosteroid levels are often elevated during stress, these findings may explain the increased levels of Th2 cytokines found in astronauts post-flight. Mice subjected to HU also have increased Th2 responses and serum glucocorticoids. We are still examining the role of GzmB in immune responses of mice subjected to HU.

3. We continue our studies on the elevation in haptoglobin in HU-treated mice. We previously found that increased serum haptoglobin levels occur after exposure to stress. With Duane Pierson and Satish Mehta ofJSC, we have extended our studies to astronaut serum. In samples taken from 20 different astronauts before and after space flight, we have found a correlation between serum haptoglobin and viral reactivation status. The ultimate goal is to develop haptoglobin as a clinical marker for evaluation of the stress status of astronauts, as well as civilians.

4. Mesenchymal stem cells (MSCs) are a unique population of multi-potent stem cells that are derived from adult bone marrow. These cells are immunosuppressive, and also seem to protect splenic and thymic lymphocytes from the deleterious effects of microgravity and radiation. We found that transfusing mice with MSCs immediately after gamma-irradiation or before hindlimb unloading significantly reduced splenocyte losses. In vitro, irradiated lymphocytes were protected from apoptosis when co-cultured with MSCs. These results indicate that MSCs provide survival signals that prevent lymphocyte apoptosis. In fact, we found several such MSC-produced cytokines, which may allow development of pharmaceutical interventions.

5. We examined the interaction between the immune and skeletal systems. Osteopontin (OPN), a bone matrix protein, has also been shown affect the immune system--as a cytokine and possibly as a transcription factor. We examined the role of OPN in HU-induced lymphocyte reduction using OPN-knockout mice, with David Denhardt and Kathryn Wang. We have reported that OPN deficiency confers resistance to HU-induced thymocyte depletion. This protective effect derives from decreased corticosteroid levels in OPN-deficient mice, indicating that OPN is important in regulating the steroid response during HU, and probably during spaceflight.

FUTURE PLANS

1. Additional particle radiation experiments to verify our previous findings and to further elucidate the molecular mechanisms responsible for apoptosis induction. We will test the effect of deficiency in both FasL and TRAIL, and examine the role of mitochondrial factors using cytochrome C-deficient mice. We will extend our findings to lower and fractionated doses of radiation, and to high-energy proton irradiation.

2. Continue to examine GzmB-regulated Th2 responses during HU of mice and its relation to glucocorticoids using steroid receptor antagonists.

3. In studies of haptoglobin as a stress biomarker, we will test additional astronaut serum samples to examine the correlation between haptoglobin levels and virus reactivation.

4. Ongoing studies of the molecular mechanisms that contribute to the cell survival/ radiation-protective effects of MSCs.

5. Examine the effects of osf-1 gene overexpression on immune parameters in mice flown in orbit, in cooperation with ASI and NASA.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	We hope to gain an understanding of the mechanisms by which stress affects immune homeostasis. With this information, it may be possible to devise dietary and lifestyle changes that would reduce the deleterious effects of stress on the immune system. Such changes may lead to an improvement in public health. We also hope to identify food supplements, such as vitamin C and seaweed extracts that reduce the health impact of stress, bringing benefit to the general population, especially those who suffer from high levels of occupational or life stress. By determining the role of cellular apoptosis in mediating the deleterious effects of stress on the immune system we may be able to aid in the development of therapies for the maintenance of immune tolerance for clinical use in organ transplant or autoimmune disease. We are studying the role of a specific subset of T lymphocytes called CD4+CD25+ regulatory T cells, which seem to be involved in the deleterious effect of stress on the immune system. Learning how these cells exert their effects may lead to potential interventions to prevent the negative health impacts of autoimmunity, cancer, and other diseases that arise from an improperly regulated immune system. It is difficult to measure the level of stress that an individual experiences for the purpose of allowing appropriate intervention. Identification of particular serum proteins whose concentration is related to the stress level of an individual could serve as a biomarker for the assessment of stress levels for medical or occupational purposes.
Task Progress:	1. Elucidate the role of Fas and endogenous opioids in the modulation of the immune system by hindlimb unloading (HU) and radiation: We have found that the synergistic loss of lymphocytes caused by gamma-radiation and HU is exerted through Fas. Our studies also revealed a role for TRAIL in these effects. We have now shown that osteopontin (OPN), a bone matrix protein, can modulate lymphocyte apoptosis during HU by reducing stress-induced-glucocorticoid production. OPN-deficient mice are resistant to HU-induced thymocyte depletion, revealing a novel function of OPN. 2. Determine the effect of HU and radiation on lymphocyte dynamics and immune responses: We found that whole-body irradiation of mice with low-dose 56Fe ion radiation (50, 100, 200 cGy) depletes splenic and thymic lymphocytes, with thymocytes being more sensitive than splenocytes. All lymphocyte subsets in spleen and thymus were affected, except for the CD4+CD25+ regulatory T cells, which were quite resistant. T cells and B cells were equally sensitive, while CD8+ T cells were more sensitive than CD4+ T cells. In the thymus, immature CD4+CD8+ (double-positive) thymocytes are exquisitely sensitive, mature single-positive cells were less sensitive, and immature double-negative cells were nearly completely resistant to these doses. These findings reveal for the first time the differential susceptibility of different lymphocyte subpopulations to particle radiation-induced deletion. This particle radiation-induced lymphocyte apoptosis was independent of Fas and TRAIL (TNF-related apoptosis-inducing ligand), most prevalent receptor-induced modes of apoptosis, since deficiency in these genes did not prevent lymphocyte losses. This is striking, since lymphocyte depletion following gamma-irradiation depends on the Fas-FasL pathway. Thus, there are distinct pathways of apoptosis induction by particle radiation compared to gamma-radiation. Furthermore, we found that particle radiation-induced lymphocyte apoptosis is caspase dependent, but does not require nitric oxide or oxygen free radicals. 3. Determine whether HU and radiation promote the development of autoimmunity: We found that unlike apoptotic cells, which promote immune tolerance, necrotic cells actually stimulate an immune response. In early studies on the development of autoimmunity following HU and concomitant irradiation, our findings as yet are inconclusive. Our molecular studies have shown that apoptotic cells cause dendritic cells (DC) to lose their ability to promote T cell production of IFN-g, while at the same time these DCs become highly sensitive to induction of nitric oxide production in response to IFN-g. The nitric oxide produced by apoptotic cell-primed DCs can in turn suppresses T cell responses. When apoptotic cells are too abundant, however, they can undergo secondary necrosis. Therefore, efficient removal of apoptotic cells is crucial in preventing autoimmunity in radiation-exposed individuals.
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